

Project Initialization and Planning Phase

Date	7 July 2026
Team ID	XXXXXX
Project Title	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

Project Proposal (Proposed Solution)

The proposed solution aims to transform the credit card eligibility assessment process using Machine Learning to improve efficiency, consistency, and decision-making. Traditional credit card application screening often involves manual verification, making the process time-consuming and prone to human error. The proposed system automates the preliminary eligibility assessment by analyzing applicant demographic, financial, and credit-related information using machine learning algorithms. The application provides instant eligibility predictions, confidence scores, and personalized recommendations through a user-friendly web interface, helping both applicants and financial institutions make informed decisions.

Section	Details
Objective	The primary objective is to develop an intelligent web-based system that predicts credit card eligibility using machine learning techniques, enabling faster, more accurate, and consistent preliminary assessments.
Scope	The project covers data preprocessing, feature engineering, machine learning model training, model evaluation, and deployment of a Flask-based web application that provides real-time credit card eligibility predictions.

Problem Statement

Section	Details
Description	Banks and financial institutions receive a large number of credit card applications every day. Manual screening is time-consuming, inconsistent, and may result in delays or incorrect decisions. Applicants also lack a reliable way to estimate their eligibility before submitting an application.

Impact	Automating the eligibility assessment process reduces manual effort, improves consistency in decision-making, speeds up application screening, enhances customer experience, and helps financial institutions identify high-risk applicants early.
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Proposed Solution

Section	Details
Approach	Build a machine learning-based web application that analyzes applicant information, preprocesses the data, and predicts credit card eligibility using a trained XGBoost classification model. The system provides real-time predictions through a Flask web interface.
Key Features	- Machine learning-based credit card eligibility prediction. - Automated data preprocessing and feature engineering. - Comparison of multiple classification algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost). - Real-time eligibility prediction with confidence score and risk level. - Personalized recommendations for applicants. - Responsive Flask web application. - Cloud deployment using Render.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i3/i5/i7 Processor or equivalent
	Memory	Minimum 8 GB RAM
	Storage	Minimum 20 GB Free Disk Space
Software	Framework	Flask
	Programming Language	Python 3.10+
	Libraries	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib, OpenPyXL
	Development Environment	Visual Studio Code / PyCharm
	Version Control	Git & GitHub
	Deployment Platform	Render with Gunicorn

Data	Dataset	CreditCardApproval_Selected_Features.csv
	Source	Kaggle (Credit Card Approval Dataset)
	Format	CSV
	Records	Approximately 307,000+ applicant records
	Features	Selected demographic, financial, employment, housing, and credit inquiry attributes used for machine learning prediction.